



Islamic University of Technology
Department of Computer Science and Engineering

Lab 11: Randomized Algorithms

CSE 4404: Algorithms Lab
Summer 2023-24

Task A. Estimating PI

Time Limit: 1 second | Memory Limit: 512 MB

You are asked to estimate the value of the mathematical constant π using the Monte Carlo method.

The idea is simple: if we randomly place points inside a square of side length 1, the probability that a point lies inside the quarter circle of radius 1 (inscribed in the square) is $\pi/4$.

Generate n random points, your job is to count how many of them fall inside the quarter circle, and then use this to estimate π .

Input Format

This problem has no input.

Output Format

Print a real number — your estimate of π .

Notes

A point (x, y) lies inside the quarter circle if $x^2 + y^2 \leq 1$.

Experiment with different values of n and observe the accuracy of the estimation.

Task B. Randomized k-th Element Search

Time Limit: 1 second | Memory Limit: 512 MB

You are asked to find the k -th smallest element in an array using a randomized Las Vegas algorithm.

Recall that a Las Vegas algorithm always produces the correct result, but its running time may vary depending on random choices. In this case, you are expected to implement an algorithm that runs in expected $\Theta(n)$ time.

Input Format

The first line contains two integers n and k ($1 \leq k \leq n \leq 10^5$) — the size of the array and the index of the element to find.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($-10^9 \leq a_i \leq 10^9$) — the elements of the array.

Output Format

Print one integer — the k -th smallest element of the array.

Examples

Sample Input	Sample Output
5 3 7 2 9 4 1	4
6 6 10 10 10 10 10 10	10
8 5 8 1 6 3 7 2 9 4	6
11 6 5 1 9 2 8 3 7 4 6 10 11	6
7 4 1000000000 -500000000 700000000 -1000000000 250000000 999999999 -750000000	250000000

Notes

The array may contain duplicate values. The k -th smallest element is defined as the element that would appear at position k if the array were sorted in non-decreasing order.

Marks Distribution

Task	Marks
Task A	50%
Task B	50%

Practice Problems

Problem	Links
Guess the Cycle Size	Codeforces: codeforces.com/contest/1729/problem/E Vjudge: vjudge.net/problem/CodeForces-1729E